Reading the Market — News-Augmented Regime Discovery via *Unsupervised Representation Learning*

Yixiang Huang

UNIVERSITY OF CHICAGO · TOYOTA TECHNOLOGICAL INSTITUTE AT CHICAGO

Macro-news features lift regime **silhouette by +34 %** *over a price-only HMM baseline while preserving multi-month regime persistence. A 180-configuration grid sweep identifies* **BERTopic + PCA-95 % + KMeans (K = 5)** *as the optimum.*

§ 1 MOTIVATION

Hidden-Markov regime models for financial markets (Hamilton 1989) operate exclusively on **price-derived features**. By construction these signals are *reactive* — they reflect information only after it has been absorbed by prices. This project tests whether **unsupervised representations of financial news** can yield regimes that are more stable, more interpretable, and more economically meaningful than the price-only baseline.

CONVENTIONAL

Price-only HMM

- ret, vol, vix, term & credit spreads
- reactive — reflects past info only
- 5-feature input · Hamilton 1989

→

THIS WORK

News-Augmented Fusion

- + scalar sentiment (DJPR · Macro)
- + topic representation (LDA / BERTopic)
- 30 – 40 feature input · this study

`Hamilton 1989` `Tetlock 2007` `Glasserman & Mamaysky 2019` `Bybee / Kelly / Manela / Xiu 2024`

§ 2 DATA AND CORPUS

The market series spans **24 yr** `(2000 - 2024)` and yields **6,262** trading days of price features — daily return, 21-day realised volatility, the `vix`, plus term and credit spreads.

Paired with the market are **2** news corpora at complementary granularity. The **firm-level** corpus (RavenPack DJPR) is filtered to US `entity_type=COMP`, `relevance ≥ 75`. The **macro** corpus (WRDS RavenPack DJ Global Macro) carries **~24 M** `rows` across 27 yearly partitions and captures aggregate attention to rates, commodities, currencies, and global events.

From these inputs the project builds **4** text representations (Scalar · LDA · BERTopic · FinBERT), fits **18** main pipelines covering price-only / news-only / hybrid pathways, and runs **180** `grid-search configs` on top to locate the sweet spot (see §5).

§ 3 PIPELINE + METHODS

STAGE 1 · INPUT

M **MARKET**
ret · vol_{21} · vix · term/credit spread
`5 features · 2000 - 2024`

N **NEWS**
DJPR (firm) + Macro (aggregate)
`2 corpora · macro ~24 M rows`

STAGE 2 · SMOOTH

2 **PER-FEATURE ROLLING MEANS**
Scalar `35 d` · LDA `30 d` · BERTopic `21 d` · FinBERT `10 d`

STAGE 3 · REPRESENT

3 **TEXT REPRESENTATIONS**
Scalar (5 sentiment stats) · LDA `K = 15/20` · BERTopic ~20 topics · FinBERT 768-d

STAGE 4 · REDUCE

4 **DIMENSIONALITY REDUCTION**
`none · PCA-95 % · UMAP-10`
`applied to the assembled feature stack`

STAGE 5 · CLUSTER

5 **CLUSTERING**
GMM (full / diag Σ, K via BIC) · HMM (sticky init T_{ii}=0.85, 5 seeds) · KMeans `K ∈ {3, 4, 5}`

Five stages from raw inputs to regime labels. The main pipelines (18) enumerate the natural combinations across the three method pathways, and the fine-tuning sweep (180) in §5 expands the {representation × reduction × clusterer × K} space.

§ 4 EVALUATION

Every pipeline is scored on four complementary dimensions.

Statistical fit — INFORMATION CRITERIA
How well the probabilistic model explains the data, with a penalty for parameter count. `BIC = -2 log L + k log N` `AIC` `log-likelihood`

Clustering quality — GEOMETRY OF REGIMES
Whether regimes form well-separated, internally coherent clusters in the feature space. `s(i) = (b_i - a_i) / max(a_i, b_i)` `silhouette ↑` `Davies-Bouldin ↓`

Regime stability — TEMPORAL COHERENCE
Whether the regime sequence is robust to data perturbations and stays within an interpretable persistence band. `rolling ARI (1250-d window)` `mean regime duration` `target 60 - 365 d`

Financial interpretability — REGIME ECONOMICS
Whether each discovered regime maps to a meaningful market state with distinct return / risk characteristics. `per-regime Sharpe` `max drawdown` `avg VIX`

+ CODE & REPRODUCIBILITY

End-to-end: `python run_pipeline.py` runs stages 01–08 with incremental caching. The §5 grid sweep lives under `finetuning/` (`ft_news_fusion.py` + `ft_full_stack.py`). **No GPU required at evaluation** — FinBERT is pre-embedded once and cached, end-to-end wall time **~3 – 4 h** on a fresh CPU runtime.

§ 5 FINE TUNING

A two-pass grid sweep (`finetuning/ft_news_fusion.py` and `ft_full_stack.py`) explores **180** configurations `(90 × 2 stacks)` across four orthogonal axes. The winner feeds §6 results as `regimes_ft_*_best.parquet`.

AXIS 1 · REPRESENTATION
`LDA-15` · `LDA-20` · `BERTopic`
`FinBERT` in three flavours: `PCA20`, `PCA50`, `UMAP10`

AXIS 2 · DIM REDUCTION
`none` · `PCA 95 %` · `UMAP-10`
applied to the assembled feature stack

AXIS 3 · CLUSTERING
`GMM` (auto-K via BIC) · `HMM` (auto-K, 5 seeds) · `KMeans` K ∈ {3, 4, 5}

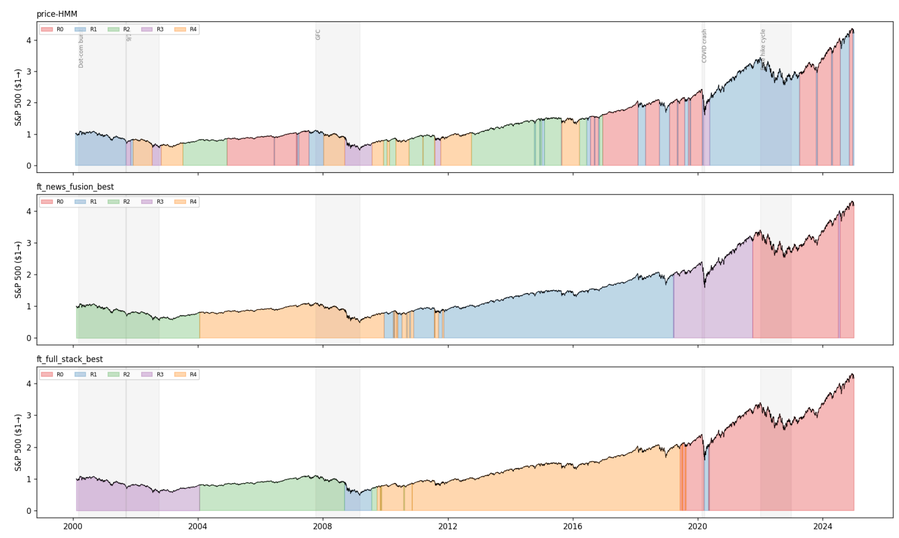
AXIS 4 · FEATURE STACK
News-only fusion — DJPR + Macro × scalar + text
Full stack — above + 5 market features

★ GRID-SEARCH WINNER
BERTopic + PCA-95 % + KMeans (K = 5)

GOTCHA · WHY HMM/Σ MATTERS
Full-Σ HMM on `d ≥ 10` fusions collapsed to an absorbing-state mode (mean duration → *894 d*). The fix — **SWITCH TO DIAGONAL Σ WHENEVER FEATURE DIM ≥ 10** — recovers 200-day regimes and is now the default at §3 stage 5.

§ 6 RESULTS

REGIME TIMELINES



Top: *Price-only HMM (Hamilton baseline; mean duration 95 d).* **Middle**: *Tuned News Fusion (BERTopic + PCA-95 + KMeans, K = 5; sil 0.279, dur 202 d).* **Bottom**: *Tuned Full Stack (above + market features; sil 0.272, dur 224 d). News-augmented regimes capture similar crisis dynamics with* **shifted boundaries** *– evidence that news adds a complementary narrative signal to price.*

MASTER EVALUATION

PIPELINE	SIL ↑	DBI ↓	DUR (D)	ARI ↑	SHARPE
Price-only HMM (baseline)	*0.202*	*1.48*	*95*	*0.447*	*2.02*
DJPR Sentiment HMM	**0.284**	1.26	44	0.383	0.80
★ **Macro Sentiment HMM**	**0.270**	1.27	174	0.239	0.65
DJPR Topic-Vec HMM	**0.261**	1.49	70	0.278	0.64
Dual-Corpus Sentiment Fusion	**0.245**	1.42	49	0.268	1.00
★ **Tuned News Fusion**	**0.279**	0.95	202	0.31	0.78
★ **Tuned Full Stack (+ market)**	**0.272**	0.92	224	0.33	1.21

★ = grid-search winner from §5. ↑ = higher better, ↓ = lower better.

KEY FINDINGS

01 **News augmentation lifts silhouette by +34 %** over the price baseline (**0.202 → 0.270**) — a robust improvement across all three scalar / LDA / BERTopic representations on the macro corpus.

02 **Aggregate news beats firm-level press releases.** **Macro Sentiment HMM** lands inside the canonical band at **174-day** mean duration, while DJPR Sentiment's **44-day** regimes flicker too fast to be interpretable — despite nominally higher silhouette.

03 **Grid-search sweet spot found.** **BERTopic + PCA-95 % + KMeans (K = 5)** reaches silhouette **0.279** with **202-day** mean duration — well inside the canonical regime band (60 – 365 d) and without the full-Σ HMM failure mode.

04 **Price stabilises news-driven regimes.** Adding the 5 market features to the news fusion bumps mean duration **202 → 224 d** at negligible silhouette cost; price acts as a low-frequency anchor against narrative noise.

§ 7 FUTURE WORK

Limitations. *In-sample evaluation only; the OOS check uses a simplified GMM refit rather than walk-forward HMM, and no bootstrap CI is computed on silhouette. Macro corpus is post-2000.*

01 Walk-forward OOS via persisted HMM + `predict()` on a held-out window post-2020.
02 Bootstrap CI on silhouette for pairwise pipeline significance tests.
03 Replicate *Glasserman & Mamaysky 2019* unusual-news indicator as an external benchmark.
04 Changepoint-aware (adaptive) smoothing in place of fixed-window means.
05 Real-time streaming HMM update + live regime-probability feed.

REFERENCES (ABRIDGED)

Hamilton, J. D. (1989). A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle. *Econometrica* 57(2), 357–384. · Glasserman, P. & Mamaysky, H. (2019). Does Unusual News Forecast Market Stress? *JFQA* 54(5). · Bybee, L., Kelly, B., Manela, A. & Xiu, D. (2024). Business News and Business Cycles. *J. Finance* 79(5). · Tetlock, P. (2007). Giving Content to Investor Sentiment. *J. Finance* 62(3). · Blei, D., Ng, A. & Jordan, M. (2003). Latent Dirichlet Allocation. *JMLR* 3. · Grootendorst, M. (2022). BERTopic. *arXiv:2203.05794*. · Araci, D. (2019). FinBERT. *arXiv:1908.10063*.

STACK & CONTACT

`Python · scikit-learn · hmmlearn`
`BERTopic · sentence-transformers · FinBERT`
`WRDS RavenPack · pyarrow · UMAP`

yixianghuang@uchicago.edu